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Sensing Plant Photosynthesis Using Solar-Induced Chlorophyll Fluorescence: From Chloroplasts to the Globe

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Abstract

Photosynthesis is the fundamental biological process that introduced oxygen into Earth's atmosphere and continues to power life, from the earliest single-celled organisms to entire global ecosystems. Yet, measuring photosynthesis across scales has been challenging because traditional techniques have not transcended scales. The emergence of remote-sensing techniques to measure solar-induced chlorophyll fluorescence (SIF) provides a unique approach to estimate photosynthesis across spatiotemporal scales, representing a new age for optical remote sensing to study photosynthesis and shaping the decades of satellite SIF research. Here, focusing on spatiotemporal scales, we review the mechanisms that drive the relationship between SIF and photosynthesis. Remotely sensed SIF is modulated by biological drivers, environmental drivers, the interaction between biological and environmental drivers, and the viewing geometry. Studying fluorescence at small scales provides the ecophysiological understanding needed to disentangle the biological and environmental drivers of SIF at larger scales. Leveraging progress in satellite SIF, future research should focus on cross-scale mechanistic understanding of the drivers of SIF and using SIF as a metric for plant function beyond photosynthesis.

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1. INTRODUCTION

Fluorescence, the emission of radiation following the absorption of shorter-wavelength radiation, is a ubiquitous phenomenon in nature (31). Many objects fluoresce with the right light source: minerals, corals, scorpions, phytoplankton, and leaves, for example. Chlorophyll molecules absorb radiation in the visible wavelengths and emit fluorescence in a longer wavelength range (640–850 nm) (112). In a dark environment (e.g., indoors or in a leaf chamber), fluorescence can be readily observed by sensitive instruments and appropriate optical filters, even though most photons are emitted beyond the visible spectral range. In nature, this weak fluorescence signal caused by the absorption of sunlight by chlorophyll was challenging to observe without filters because of the strong background radiation. With the advancement of technology and algorithms, solar-induced chlorophyll fluorescence (SIF), the fluorescence emission when chlorophyll molecules absorb sunlight, can now be detected by satellite sensors at the global scale, leveraging very-high-resolution spectrometers (37, 51, 72). SIF has been used as a proxy for vegetation photosynthesis (13) and numerous applications related to crop yield prediction, phenotyping, and stress monitoring (48, 52, 150, 167). This exciting development opens up the opportunity to bridge the long tradition of laboratory-scale studies of fluorescence emission dynamics associated with leaf biochemistry to global change ecology (7, 13, 36, 93, 102, 121, 137).

At the global scale, photosynthesis provides more than 99% of the biochemical energy that the biosphere uses, harnessing approximately 3% of the solar irradiance at Earth's surface each year (60). The process provides the food used by nearly all microbes and animals, including humans (36), and is the largest CO_2 flux in the global carbon cycle, which is largely offset by ecosystem respiration (38). Global annual terrestrial photosynthesis [123±8 Pg C/yr (11)], estimated from modeled products derived from eddy covariance measurements sampling a representative fraction of the land surface, is about 11 times greater than anthropogenic CO_2 emissions (10.8±0.9 Pg C/yr) (38). The uncertainty in global terrestrial photosynthesis is almost three times the land carbon uptake (3.2±0.9 Pg C/yr) (38). A key challenge in understanding photosynthesis in a changing global environment is to measure the process at geographically significant scales. While gas exchange systems are convenient for laboratory studies, they require plant tissue to be enclosed in a chamber. Enclosure becomes difficult to impossible when measuring photosynthesis at a larger scale, such as a crop or forest canopy (27). The global network of eddy covariance sites has provided estimates of gross primary productivity (GPP) in diverse ecosystems around the world, greatly contributing to our ability to understand environmental constraints on canopy photosynthesis (8). While eddy covariance systems have excellent temporal resolution, they have limited spatial extent. Global

Eddy covariance: a micrometeorological approach to measure the exchange of CO_2 and H_2O between land and atmosphere that can be used to estimate photosynthesis

Gross primary productivity (GPP): apparent photosynthesis (gross photosynthesis—photorespiration) at the ecosystem scale

and spatially resolved photosynthesis measurements are needed to understand the global carbon cycle and how photosynthesis and food security will respond to global environmental changes.

Remote sensing has emerged as a powerful tool for addressing this challenge because of its global coverage (127). One key feature of terrestrial plant canopies is that chlorophyll is embedded in a highly scattering environment (within and among leaves). Consequently, photons ricochet within the canopy and interact with scattering elements and chloroplasts. Some of these scattered photons escape from the canopy, where remote-sensing instruments can detect them. Because red photons are absorbed by chlorophyll molecules while near-infrared (NIR) photons are mainly scattered by plant materials, comparison of the spectral composition of the escaping photons to that of those entering the canopy provides information on the flux of light absorbed by the canopy and, thus, its photosynthetic activities (130). Since the 1970s, this has been the basis of the classic remote-sensing approach that measures canopy greenness via Vegetation Indices [VIs, e.g., the Normalized Difference Vegetation Index (NDVI)] (139).

Vegetation Indices (VIs): indices that are mathematical transformations of spectral reflectance from individual bands

Normalized Difference Vegetation Index (NDVI): NDVI is calculated as (NIR − R)/(NIR + R); NIR and R are the reflectance values (0–1) in the near-infrared and red wavelengths

Recently, SIF has emerged as a promising method for estimating photosynthesis more directly. Instead of using the reflected solar radiation from the vegetation canopy, the SIF approach leverages the emitted fluorescence photons in narrow bands in the red and NIR wavelengths. As a significant departure from VIs, SIF has a clear physical meaning for photosynthesis function and plant physiology. While plant biologists have widely used chlorophyll *a* fluorescence at the leaf level to understand mechanisms that drive photosynthesis and responses to environmental stress (7, 25, 43, 78, 149), scaling these measurements to larger areas has historically been challenging due to the tiny fluorescence signal being masked by intense reflected sunlight. The development of ultrahigh-spectral-resolution spectroscopy has enabled remote-sensing instruments, whether spaceborne, airborne, or ground-based, to isolate SIF signals. As a result, recent studies have demonstrated the power of SIF in tracking photosynthesis at the global scale (37, 52, 72).

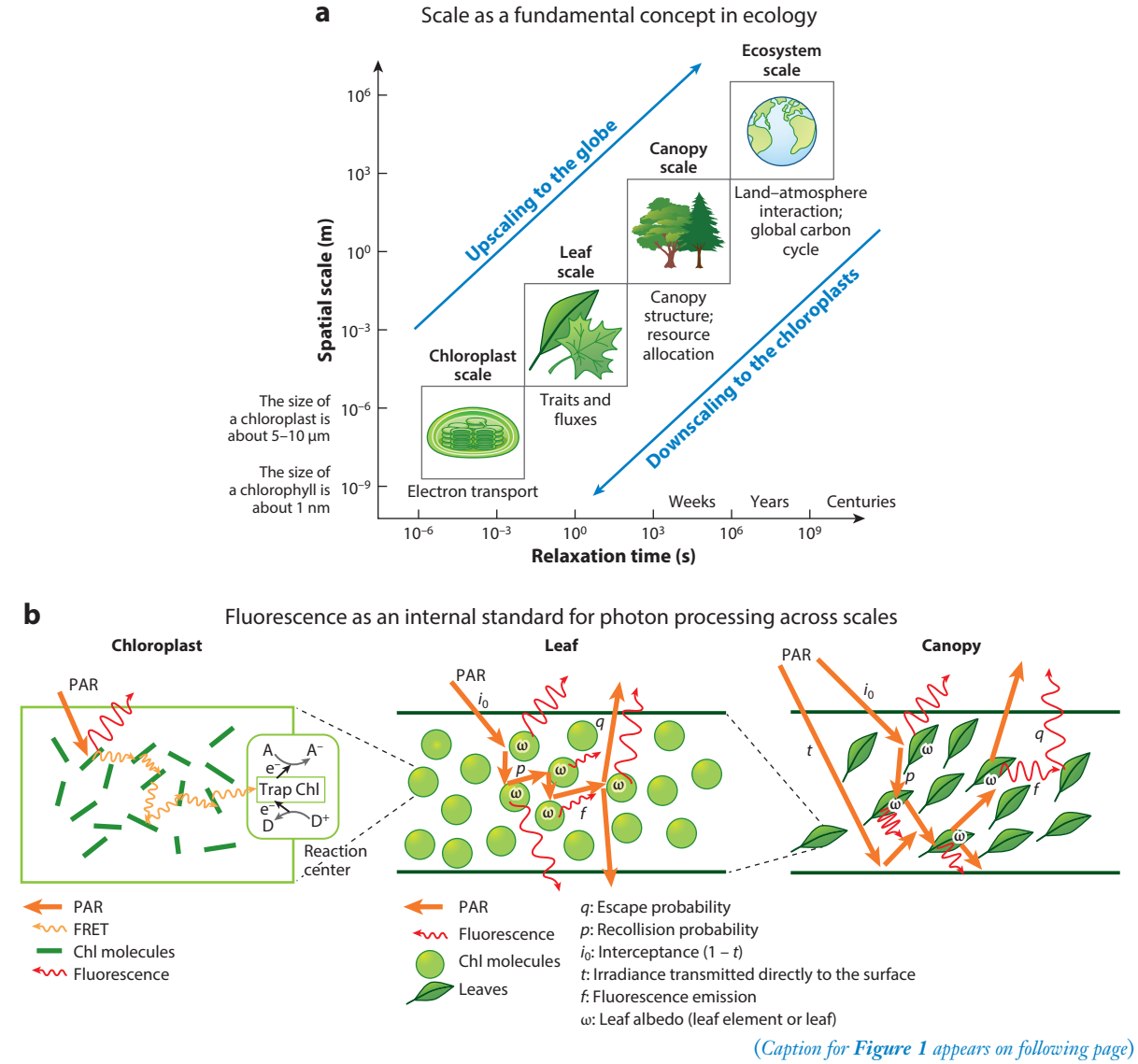
Utilizing SIF to improve the understanding of ecosystem- and global-scale photosynthesis requires a deeper understanding of the biological mechanisms that drive the spatiotemporal variability of SIF (**Figure 1**). Just like photosynthesis, the observed spatiotemporal patterns of SIF are products of processes at multiple scales, from the chloroplast scale to the canopy, ecosystem, and global scales. Indeed, measurements on single leaves often do not capture the variability within a highly heterogeneous canopy. Leaves within a canopy do not function solely as isolated individuals; instead, they interact with other leaves in the canopy and collectively contribute to canopy photosynthesis (103). Thus, the structure of plants and plant canopies, including the spatial arrangement of leaves, is crucial to deciphering the SIF–photosynthesis relationship.

In this review, we aim to provide a critical perspective on how understanding the ecophysiological mechanisms of fluorescence emission and scaling of biological processes can help scientists work at different scales (chloroplast, leaf, canopy, ecosystem, and globe). While some studies have shown a linear relationship between SIF and photosynthesis, others have found this relationship to be more complicated (37, 57, 72, 163). A key question that we would like to address is, At what scales does a linear relationship emerge between SIF and photosynthesis, and what are the factors that control the degree of linearity in the SIF–photosynthesis relationship at different scales? We emphasize the need for a mechanistic understanding of SIF observations at large scales, which can only be accomplished by applying knowledge gained at finer scales, requiring collaboration between plant scientists and remote-sensing scientists. In this review, we first present the basis of chlorophyll fluorescence biochemistry (7, 13, 121), focusing on the steady-state yields of fluorescence and photochemistry. We then discuss the central theme of SIF research, which is how SIF and photosynthesis are related to each other at various spatiotemporal scales. We provide a brief review of the SIF observations and modeling approaches. We end the review with key questions to be answered in the near future.

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2. THE FOUNDATION OF CHLOROPHYLL FLUORESCENCE BIOCHEMISTRY

Chlorophyll *a* is used in photochemistry in all vascular plants (13). The energy of a photon can be trapped and held in a metastable state for a few picoseconds as an excited electronic state of chlorophyll, often referred to as an exciton, which can be transferred among closely packed chlorophylls or other pigments (**Figure 1**). One possible fate is for these excitons to power a charge separation between a special pair of chlorophylls, initiating a chain of redox reactions that eventually split water to produce O_2, and generate nicotinamide adenine dinucleotide phosphate (NADPH) and adenosine triphosphate (ATP), enabling the fixation of CO_2 to produce sugars. This is termed photochemical quenching because the absorbed energy is directed toward the energy-conserving



(*Caption for* ***Figure 1*** *appears on following page*)

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Figure 1 (*Figure appears on preceding page*)

Scale is a fundamental concept in understanding the relationship between photosynthesis and fluorescence. Fluorescence serves as an internal standard for photon processing at the chloroplast, leaf, and canopy scales. (*a*) The relaxation time (in seconds) is the time it takes for a system to return to a steady state after perturbation. The two blue arrows represent our recommendation that cross-scale understanding and discussion are necessary to move the field forward. Plant biologists have been working on fluorescence at the molecule to leaf levels for decades, mostly focusing on active fluorescence, specifically the pulse amplitude modulation technique. The applications of steady-state fluorescence at the canopy level are less well-studied. Vegetation structure significantly impacts photosynthesis and fluorescence. Linking photosystem-level mechanisms to large-scale observations requires understanding the biology and radiative transfer of plant canopies. Studies at the ecosystem scale can benefit from a more mechanistic understanding of fluorescence at the leaf scale. The relationship between solar-induced chlorophyll fluorescence and gross primary productivity at a specific spatial scale can also change with the temporal scale. (*b*) Radiative transfer within a chloroplast, a leaf, and a canopy. (*Left*) At the chloroplast level, a similar transfer of energy to that observed at larger scales can be seen among chlorophyll molecules via fluorescence resonance energy transfer, a nonradiative process in which energy trapped by a chlorophyll molecule is transferred to another. Eventually, the energy can be transferred to the reaction center and trapped by specialized chlorophyll molecules (P680 and P700 in plants, algae, and cyanobacteria). Here, light energy is converted into chemical potential energy by transferring electrons from the Donor (D+) to the Acceptor (A−) in the photosynthetic reaction centers (45). (*Right*) At the canopy scale, the incoming photosynthetically active radiation (PAR) is either intercepted by the canopy (characterized by the photon interception ratio i_0, 0–1, representing the photon interception by the canopy) or directly transmitted through the canopy to the ground surface (characterized by the transmittance t, 0–1). The intercepted PAR is absorbed, reflected, or scattered by plant materials (including leaves and woody materials). Fluorescence is emitted (f) as one of the pathways after PAR is absorbed by chlorophyll molecules. The probability of the scattered PAR being intercepted by a leaf is p, the recollision probability. ω is the leaf albedo (the sum of leaf transmittance and reflectance). (*Center*) Similar radiative transfer processes are illustrated for a single leaf (92). Fluorescence emission can be reabsorbed before it escapes the canopy or the leaf tissue. In addition, fluorescence emission is emitted in all directions, and sensors normally pick up the emission from the viewing direction. For a comprehensive review of radiative transfer models, see Reference 64. Panel *a* adapted with permission from Reference 110; copyright 2014 Annual Reviews, and Reference 111; copyright 1988 CSIRO Publishing.

formation of photochemical bonds. Nonphotochemical mechanisms of de-excitation compete with photochemistry, leading to dissipation of the absorbed energy in the form of heat. These processes are collectively referred to as nonphotochemical quenching (NPQ). Under normal growth light conditions, energy dissipation by NPQ is overall much greater than energy capture by photosynthesis and energy dissipation by fluorescence combined. Another fate of the exciton is to be reemitted as light that generally has a slightly longer wavelength and lower energy—fluorescence.

Intrinsic lifetime of fluorescence (τ_0): also the natural lifetime of chlorophyll fluorescence, it is the time it takes fluorescence to decay by 1/e

Rate constant (K): quantifies the relationship between the rate of a chemical reaction and the concentration of the reactant; for chlorophyll fluorescence, K_F is 0.05 ns^{-1}

Fluorescence yield (Φ_F): the ratio between the number of fluorescence photons and the number of photons absorbed by chlorophyll molecules

A valuable property of chlorophyll fluorescence is that the probability of the exciton escaping as fluorescence in a given time interval is an intrinsic property of the chlorophyll molecule, which is inversely proportional to the in vitro intrinsic lifetime of fluorescence (τ_0) ($\tau_0 = 1/K_F$, ns) where K_F is the rate constant for fluorescence. The actual lifetime of chlorophyll fluorescence is modulated by various processes that compete with fluorescence within the chloroplast. Therefore, fluorescence can serve as a built-in probe of the activity of the excitation mechanisms occurring in photosynthetic systems (121). When other pathways [with rate constant(s) (K_{others})] are available for de-excitation, the apparent fluorescence lifetime will be shorter [$\tau_F = 1/(K_F + K_{others})$], and the fluorescence yield (Φ_F) will be lower [$\Phi_F = K_F \cdot \tau_F = K_F/(K_F + K_{others})$]. Since K_F is the same constant for chlorophyll molecules in all organisms that perform oxygenic photosynthesis, the changes in τ_F and Φ_F provide a universal and quantitative probe for the rates of photochemical quenching and NPQ. To introduce the current approaches for this, we start with their historical foundation.

Scientists' interest in chlorophyll fluorescence dates back to the 1800s. Some of the earliest qualitative descriptions of the Stokes shift, a phenomenon in which the emitted radiation has a longer wavelength than the absorbed radiation, come from von Goethe's book *Theory of Colours*, describing Thomas Johann Seebeck's experiments with minerals (145). Brewster (16) made the first documented discovery of chlorophyll fluorescence by shining bright sunlight through a chlorophyll solution extracted from the leaves of cherry laurel (*Prunus laurocerasus*): "[I] was

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surprised to observe that its color was a brilliant red complementary to the green" (p. 542). This represents a qualitative discovery of the Stokes shift in chlorophyll. Stokes (134) coined the term fluorescence, referring to the mineral fluorspar, though Edmond Becquerel discovered the same phenomenon nine years earlier (10). In 1916, Einstein proposed three coefficients (i.e., the Einstein coefficients) that describe the probability of the absorption and emission of photons, which is key to understanding the process of fluorescence emission (31).

Fluorescence radiance: the fluorescence energy per unit area per unit wavelength per solid angle ($W/m^2/nm/sr$)

Scientists started to relate fluorescence and photosynthesis in the early twentieth century (112). In the 1930s, Kautsky (74) was the first to note the rapid increase of chlorophyll fluorescence followed by a gradual decrease in dark-adapted leaves exposed to light, attributing this phenomenon qualitatively to the concurrent changes in CO_2 assimilation measured in another earlier article. In 1940, McAlister & Myers (98) measured hundreds of time-resolved curves of CO_2 uptake and fluorescence in the first few minutes after dark-adapted leaves were exposed to light. This was the first time that a positive relationship between steady-state fluorescence and photosynthesis had been documented. Yet this positive relationship was not widely utilized until the recent development of SIF.

In the 1960s, Duysens et al. (29) and Tagawa et al. (138) revealed that the chlorophyll pigments emitting fluorescence were organized into two types of pigment–protein complexes that mediated different photochemical reactions. Today, these complexes are known as photosystem I (PSI) and PSII, and the major energy transduction pathways they mediate are known as linear and cyclic electron flows.

The key to using fluorescence as a quantitative probe of the inner workings of the electron transport system was the invention of the pulse amplitude modulation (PAM) technique in the period from the 1960s to the 1980s. As Krause & Weis (81) later put it, "Chlorophyll *a* fluorescence emitted by green plants reflects photosynthetic activities in a complex manner" (p. 314), and the PAM technique has enabled detailed studies of photochemical quenching and NPQ (**Figure 2**). The key to the PAM technique was the introduction of modulated light that allows for fluorescence yield measurements independent of background light (actinic light). The PAM technique has led to breakthroughs in understanding the biological and environmental factors controlling photosynthesis in intact leaves (23, 30, 42, 57, 78, 129, 149). Due to its importance, in the next section, we provide an overview of PAM's similarities and differences to SIF and discuss commonly asked questions.

3. ACTIVE VERSUS PASSIVE: DIFFERENCES AND SIMILARITIES BETWEEN PULSE AMPLITUDE MODULATION FLUORESCENCE AND SOLAR-INDUCED CHLOROPHYLL FLUORESCENCE

3.1. How Do Active and Passive Fluorometry Differ?

PAM fluorescence is an active approach, relying on a controlled light source to induce fluorescence emission and filters to isolate fluorescence signals; SIF, on the other hand, is a passive approach based on fluorescence naturally induced by sunlight. Unlike PAM, no optical filters are available for SIF retrieval, so we must rely on naturally occurring filters such as the Fraunhofer lines and the O_2 absorption features (**Figure 2**). SIF is the absolute amount of fluorescence emitted at the steady state ($W \cdot m^{-2} \cdot nm^{-1} \cdot sr^{-1}$). By contrast, PAM fluorescence is typically evaluated as the relative amount of fluorescence emitted under different conditions (i.e., steady-state as well as minimum and maximum). These fluorescence metrics are then combined and reported in dimensionless ratios. Understanding that the primary SIF and PAM measurements are both fluorescence radiances and that they are then processed to additionally provide estimates of fluorescence yields

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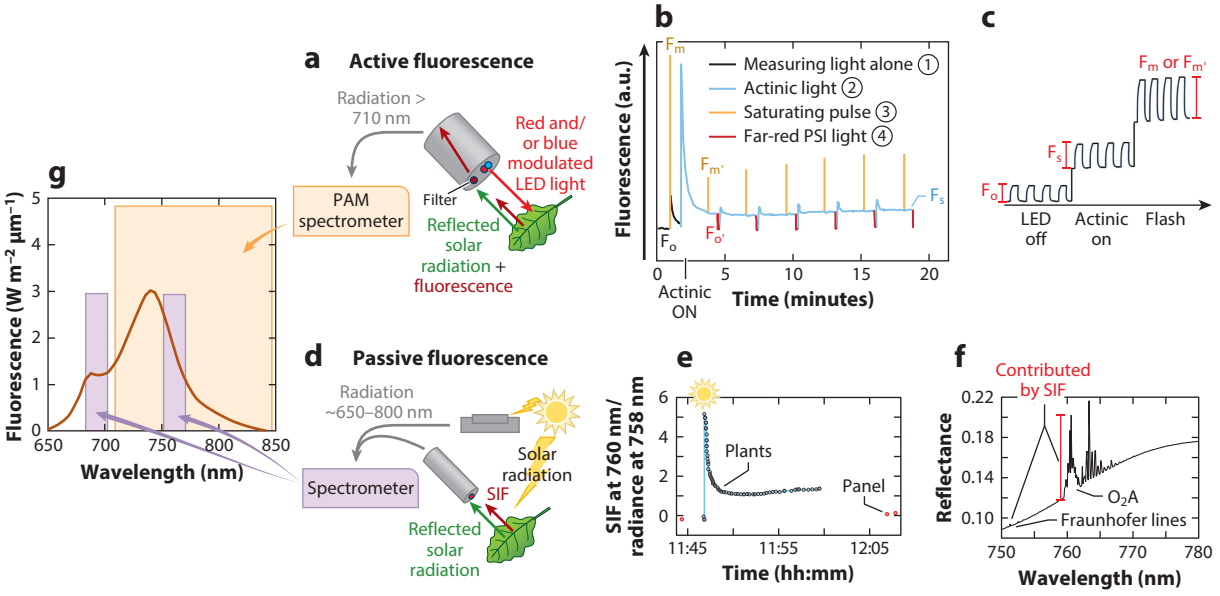


Figure 2

Main similarities and differences between active and passive fluorescence. (*a*) Active fluorescence approaches use LEDs or lasers to deliver high-frequency pulses of light to a leaf, typically in red- or blue-wavelength regions. The returning signal goes through a long-pass filter so that fluorescence radiation goes through while the blue or red measuring light is blocked. (*b*) In a typical PAM protocol, total fluorescence emission is measured under four different light conditions: (①) actinic light is off, (②) actinic light is on, (③) a brief saturation flash is applied, and (④) a brief dark pulse with far-red light is applied. Fluorescence signals induced by these pulses are denoted as F_o and F_m (minimum and maximum in dark), F_o' and F_m' (minimum and maximum in light), and F_s (steady-state in light). Panel adapted with permission from Reference 7a; copyright Society for Experimental Biology 2004, all rights reserved. (*c*) In each condition, the difference in fluorescence measured between the on and off of the constant measuring light is the useful signal. (*d*) Passive fluorescence measures the fluorescence induced by sunlight. Instruments measure the incoming solar radiation, the reflected solar radiation, and SIF. (*e*) In the field, a similar drastic increase in SIF (in this case, SIF at 760 nm/Radiance at 758 nm, proportional to the SIF yield, which is conceptually similar to fluorescence yields measured by PAM in panel *b*) was observed after a dark-adapted vegetation canopy was exposed to sunlight at noon. The panel is a nonfluorescent target and thus has a yield close to zero. Graph provided by A. Kornfeld & J. Berry. (*f*) The reflectance is the ratio of the spectra of light coming from a leaf to that of the incident sunlight. The spikes in this ratio are at wavelengths where sunlight is depleted due to absorption by O_2 at 760 nm or Fraunhofer lines near 750 nm (36). The presence of fluorescence from the leaf results in an increased apparent reflectance in these bands. Because these features occur in very narrow wave bands, very-high-resolution spectrometers (<0.5 nm) with a high signal-to-noise ratio are required. These can be located on towers above the canopy or satellites. Panel simulated with SCOPE 2.1 (160). (*g*) SIF can only be retrieved at specific wavelengths corresponding to these spectral features. The orange box indicates the wavelength range of a PAM spectrometer. The two purple boxes (around 670 nm and 759 nm) are the retrieval windows of most of the SIF algorithms. This is relevant because fluorescence is a mixture of light from PSI and PSII, which have different spectra and different sensitivity to conditions (13). Abbreviations: F_m, maximum fluorescence signal in dark; F_m', maximum fluorescence signal in light; F_o, minimum fluorescence signal in dark; F_o', minimum fluorescence signal in light; F_s, steady-state fluorescence signal in light; LED, light-emitting diode; PAM, pulse amplitude modulation; PS, photosystem; SCOPE, Soil Canopy Observation of Photosynthesis and Energy Fluxes; SIF, solar-induced chlorophyll fluorescence.

is important. While fluorescence radiance can vary by an order of magnitude or larger during a day or with PAM, steady-state fluorescence yield varies much less. Yet, the latter also provides critical information about the status of the photosynthetic apparatus. In addition, the retrieval approach differs between PAM and SIF: while the former relies on the temporal differences in fluorescence emissions induced by an active light source during a brief time interval with the help from optical filters, the latter exploits the spectral differences between the absorption bands and the neighboring wavelengths along the solar spectrum, relying on these naturally occurring filters (36). In many

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cases, the fluorescence radiance derived in many SIF studies is for one wavelength (e.g., 758 nm or 678 nm), whereas F and Φ_F in PAM integrate fluorescence emission across a broader range (e.g., >710 nm). To relate the two, we need to consider these spectral differences, the flux of absorbed photons, and the reabsorption of fluorescence within the canopy and leaves.

3.2. How Does a Pulse Amplitude Modulation Fluorometer Work?

A PAM fluorometer measures fluorescence radiance under a modulated measuring light that goes on and off in a few microseconds. The fundamental equation to derive these parameters is simple and based on the temporal differences in fluorescence emission between the on and off periods:

$$F = F(t_1) - F(t_0). \quad 1.$$

$F(t_1)$ is the fluorescence flux measured when the modulated (AC) light is on. $F(t_0)$ is the fluorescence flux measured when the modulated light is off, and it is insensitive to fluorescence that may be induced by a steady (DC) light. The difference between t_1 and t_0 is a few milliseconds, enabled by ultrafast electronic modulating systems, and permits isolation of the fluorescence radiance induced by the measuring light F. In a typical PAM protocol, this approach is used to evaluate F_o and F_m (minimum and maximum fluorescence in the absence of actinic light), F_o' and F_m' (minimum and maximum fluorescence in the presence of an intense pulse of DC light), and F_s (steady-state fluorescence in the presence of actinic light). These five primary measurements are then usually combined into a number of dimensionless ratios and/or indices, which are discussed in Section 4.

3.3. How Does a Solar-Induced Chlorophyll Fluorescence Spectrometer Work?

Unlike PAM fluorescence, SIF is retrieved based on the spectral differences between the absorption bands and the neighboring wavelengths (36, 37, 72, 99) (**Figure 2**). These algorithms, despite having different mathematical forms, share a basis for the retrieval:

$$L(\lambda) = (E(\lambda)^*\rho(\lambda)/\pi + \mathrm{SIF}(\lambda))^*T(\lambda). \quad 2.$$

The radiation received by the sensor [$L(\lambda)$] at a specific wavelength (λ) is the sum of the reflected radiation (the product of incoming radiation, E, and surface reflectance, ρ, divided by π) and the emitted fluorescence in the form of SIF, multiplied by the atmospheric transmittance $T(\lambda)$, which is driven by the amount and properties of the absorbers and scatterers and the distance between the sensor and the vegetation. Measuring E and L at multiple wavelengths in a wavelength region that straddles the absorption bands and neighboring bands allows for the retrieval of SIF at a specific wavelength. For more detailed expressions of this relationship, readers are referred to seminal works on the topic (36, 99, 143).

4. A MECHANISTIC FRAMEWORK TO INTERPRET PULSE AMPLITUDE MODULATION AND SOLAR-INDUCED CHLOROPHYLL FLUORESCENCE MEASUREMENTS

The way that the leaf-level fluorescence radiances and yields derived from PAM relate to those derived from SIF has been a focus of significant attention over the past decade. Before the SIF research field developed, the traditional approach to interpreting PAM measurements relied on combining the F_o, F_m, F_o', F_m', and F_s measurements in various dimensionless ratios and/or indices. In our view, some of the most useful of these are the expressions that estimate the three major components of energy partitioning at PSII: (*a*) the photochemical yield (Φ_P) of PSII, $\Phi_{P2} = 1 - F_s/F_m'$ (42); (*b*) the nonphotochemical quenching yield (Φ_N) of PSII for regulated

Absorption bands: spectral regions in which the incoming solar radiation intensity is reduced due to the absorptions of constituents in the solar atmosphere or in Earth's atmosphere

Photochemical yield (Φ_P): the ratio between the absorbed photons used for photochemistry and the number of photons absorbed by chlorophyll molecules

Nonphotochemical quenching yield (Φ_N, Φ_X, or Φ_D): the ratio between the absorbed photons dissipated as heat and the number of photons absorbed by chlorophyll molecules

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NPQ, $\Phi_{N2} = F_s \times (1/F_m' - 1/F_m)$ (58); and (*c*) the yield of PSII for constitutive NPQ and fluorescence, $\Phi_{D2} + \Phi_{F2} = F_s/F_m$ (58). Together, these expressions were originally proposed to permit analysis of the factors controlling the two regulated pathways for energy dissipation (i.e., Φ_{P2} and Φ_{N2}) and to situate these analyses within a complete quantitative description of the fate of light absorbed by PSII (i.e., $\Phi_{P2} + \Phi_{N2} + \Phi_{D2} + \Phi_{F2} = 1$). Over time, it has become clear that while these and other ratios/indices derived from PAM F_o, F_m, F_o', F_m', and F_s measurements are all qualitatively reasonable, they are also variably contaminated by fluorescence originating from PSI, and this effect presents a serious challenge for those applications seeking to rigorously quantify the fates of absorbed light (35, 44, 115). Fortuitously, the new measurements and conceptual frameworks developed by the SIF research community are opening new opportunities for quantitative interpretation of fluorescence.

With the rise of SIF research, there has been a concerted effort to move away from interpreting PAM measurements solely as dimensionless ratios and/or indices and toward interpreting them as radiances that can be related directly to SIF radiances. A key step in this direction was formulated by van der Tol et al. (141), who use PAM measurements to calibrate a function describing regulated NPQ at PSII in relation to the approach to light saturation of photosynthesis. In essence, van der Tol et al. (141) conceptualized the variation in the PAM and SIF fluorescence yields as being controlled by competition between effective rate constants for photochemical quenching and regulated NPQ (which vary based on the metabolic state of a leaf) and intrinsic rate constants for constitutive NPQ and fluorescence (which are constant across all metabolic states). By quantifying the relationship between regulated NPQ and photochemical quenching that is typically observed in PAM measurements, the function used by van der Tol et al. was a major step forward because it eliminated a free variable and made it possible to simulate the SIF yield and radiance using a photosynthesis model. Gu et al. (47) later developed an analogous approach that achieved a similar functional linkage between a photosynthesis model and SIF observations with an alternative intermediate function derived from a different analysis of PAM parameters. As these approaches have become more widely used, several limitations have emerged, including (*a*) a lack of generality due to sensitivity to the particular species and conditions used for calibration, (*b*) numeric inefficiency due to the number of independent parameters requiring calibration within the functions, and (*c*) limited accuracy due to the contributions of PSI to the total fluorescence detected with the PAM technique.

Johnson & Berry (67) and Johnson et al. (68) recently addressed these challenges by pivoting from the semiempirical approaches to a more mechanistic approach that achieves improved generality, efficiency, and accuracy in describing the photosynthesis–fluorescence relationship. At the core of this advance is the insight that the Cytochrome b_6f complex controls many of the overall dynamics of electron transport through PSI and PSII and the associated variation in fluorescence emission. This perspective provides the foundation for an improved mathematical description of the whole photosynthetic system, including a clearer view of the contributions that PSI and PSII each make to the overall leaf-level fluorescence radiances through their respective absorption cross sections as well as fluorescence yields.

In this framework (67), an energy conservation equation is used to track the fates of light absorbed (α) by a chloroplast or a leaf:

$$\alpha = \alpha_2 \cdot (\Phi_{P2} + \Phi_{N2} + \Phi_{D2} + \Phi_{F2}) + \alpha_1 \cdot (\Phi_{P1} + \Phi_{X1} + \Phi_{D1} + \Phi_{F1}), \quad 3a.$$

where the total amount of light absorbed (α) is partitioned between the absorption cross sections of PSII (α_2) and PSI (α_1), with

$$\alpha = \alpha_2 + \alpha_1, \quad 3b.$$

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and the light reaching each photosystem is partitioned between photochemistry (Φ_{P2}, Φ_{P1}), inducible NPQ (Φ_{N2}, Φ_{X1}), constitutive NPQ (Φ_{D2}, Φ_{D1}), and fluorescence (Φ_{F2}, Φ_{F1}), with

$$\Phi_{P2} + \Phi_{N2} + \Phi_{D2} + \Phi_{F2} = 1 \text{ and} \quad 3c.$$

$$\Phi_{P1} + \Phi_{X1} + \Phi_{D1} + \Phi_{F1} = 1. \quad 3d.$$

Under any condition, the fluorescence yield of PSII can be described with the expression

$$\Phi_{F2} = \phi_{F2}/(1 - \phi_{U2}), \text{ where} \quad 4a.$$

$$\phi_{F2} = {q^0}_{P680} \cdot [K_{F2}/(K_{P2} + K_{N2} + K_{D2} + K_{F2} + K_{U2})] + {q^-}_{P680} \cdot [K_{F2}/(K_{N2} + K_{D2} + K_{F2} + K_{U2})]$$

and

$$\phi_{U2} = {q^0}_{P680} \cdot [K_{U2}/(K_{P2} + K_{N2} + K_{D2} + K_{F2} + K_{U2})] + {q^-}_{P680} \cdot [K_{U2}/(K_{N2} + K_{D2} + K_{F2} + K_{U2})],$$

where the parameters are the fraction of open (${q^0}_{P680}$) versus closed (${q^-}_{P680}$) reaction centers; the degree of inducible NPQ (here, heat dissipation in the antennae bed: K_{N2}); the degree of connectivity between reaction centers (K_{U2}); and the intrinsic rate constants for photochemistry (K_{P2}), constitutive NPQ (K_{D2}), and fluorescence (K_{F2}).

The fluorescence yield of PSI remains constant, but it can also be described with a similar, albeit relatively simpler, expression:

$$\Phi_{F1} = {q^0}_{P700} \cdot [K_{F1}/(K_{P1} + K_{D1} + K_{F1})] + {q^+}_{P700} \cdot [K_{F1}/(K_{X1} + K_{D1} + K_{F1})], \quad 4b.$$

where the parameters are the fraction of open (${q^0}_{P700}$) versus closed (${q^+}_{P700}$) reaction centers and the intrinsic rate constants for photochemistry (K_{P1}), inducible NPQ (here, heat dissipation at closed reaction centers: K_{X1}), constitutive NPQ (K_{D1}), and fluorescence (K_{F1}).

In Equations 4a and 4b, the intrinsic rate constants can be calibrated from fluorescence lifetime measurements (i.e., K_{F2} and $K_{F1} = 0.05$ ns^{-1}, K_{D2} and $K_{D1} = 0.55$ ns^{-1}, $K_{P2} = 4.5$ ns^{-1}, and K_{P1} and $K_{X1} = 14.5$ ns^{-1}), which anchors the fluorescence yields and lifetimes for each photosystem at values that are representative of fully dark-adapted leaves under unstressed conditions (i.e., for PSII, $\Phi_{F2} = 1.0\%$ and $\tau_{F2} = 200$ ps at F_o and $\Phi_{F2} = 8.0\%$ and $\tau_{F2} = 1.6$ ns at F_m; for PSI, $\Phi_{F1} = 0.3\%$ and $\tau_{F1} = 66$ ps at both F_o and F_m). The remaining variables are then the states of the reaction centers of PSII (${q^-}_{P680}$, Equation 4a) and PSI (${q^+}_{P700}$, Equation 4b) and the level of regulated NPQ processes engaged in the antennae of PSII (K_{N2}, Equation 4a). Under any given condition, the values of these variables can be inferred simply by coupling Equations 3 and 4 to the remainder of the Johnson & Berry (67) model, which describes the flows through the electron transport system and carbon metabolism and the regulatory processes coordinating these two metabolism domains.

This framework creates a new opportunity for interpreting PAM- and SIF-style chlorophyll fluorescence measurements and relating them to other types of measurements and models. While the plant science and Earth science communities have traditionally interpreted photosynthetic CO_2 fixation through the Farquhar–von Caemmerer–Berry (FvCB) (32) model, this framework was formulated before PAM fluorimetry had matured, and it did not originally have a way to reproduce typical patterns of fluorescence. For example, it is commonly observed in leaf-level light response curves that (*a*) the CO_2 fixation rate initially increases with light but then saturates at high light intensities, (*b*) the fluorescence radiance continues to increase after CO_2 fixation has saturated, and (*c*) the steady-state fluorescence yield exhibits a skewed hump-shaped curve with a long tail under high light (see Section 5). The van der Tol et al. (141) model explains that the

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initial increase in the fluorescence yield is associated with the light-limited phase of photosynthesis in the FvCB model; that the peak is associated with the transition from light limitation to light saturation; and that the subsequent decrease is associated with the light-saturated phase of photosynthesis, where CO_2 fixation is stable. But it doesn't explain why these associations exist, that is, the specific mechanisms that control these patterns. This is provided by the Johnson & Berry (67) model: The fluorescence yield increases under light-limited conditions because the kinetic bottleneck at Cytochrome b_6f causes the accumulation of reduced plastoquinone and oxidized plastocyanin, progressively closing the reaction centers of PSII (q^-_{P680}, Equation 4a) and PSI (q^+_{P700}, Equation 4b), whereas under light-saturated conditions, fluorescence yield decreases because regulated NPQ processes are engaged in the antennae of PSII (K_{N2}, Equation 4a), dissipating excess absorbed light as heat. This mechanistic understanding makes it possible for the first time to mathematically describe both the closure of reaction centers at PSII and PSI and the development of regulated NPQ at PSII as internal variables that respond dynamically to the interaction between a chloroplast or leaf and its environment. With this capability, SIF can be interpreted and simulated more accurately, more efficiently, and over a broader range of conditions than was possible with the earlier empirical models.

We have discussed these interactions at length here because their overall result is that the chloroplast- and leaf-level fluorescence radiance, fluorescence yield, and CO_2 fixation rate do not have a simple linear relationship. Instead, these three properties have a complex, nonlinear relationship that is structured by both photosynthetic functional traits and the environment. The richness of these relationships is both a challenge and an opportunity. On the one hand, we do not yet fully understand all the factors that structure the relationships between fluorescence radiance, fluorescence yield, and CO_2 fixation, especially as they vary between different photosynthetic organisms and environmental conditions. On the other hand, it is also clear that environmental variation in fluorescence encodes unique information about photosynthetic functional traits and that we could potentially decode this at the global scale with the right combination of observing systems and modeling frameworks. To do so successfully, we must understand how to scale our understanding of photosynthetic mechanisms, and this is the topic of the next section.

fPAR: the fraction of PAR absorbed by plants

Photosynthetically active radiation (PAR): radiative energy flux between 400 nm and 700 nm that can be used for photosynthesis by plants (in W/m^2)

Photosynthetic photon flux density (PPFD): the density of PAR photons reaching a surface per unit time (in μmol photon·m^{-2}·s^{-1})

5. WHAT CAN WE LEARN OR NOT LEARN ABOUT PHOTOSYNTHESIS FROM SOLAR-INDUCED CHLOROPHYLL FLUORESCENCE AND AT WHICH SPATIOTEMPORAL SCALE(S)?

To answer these questions, we start with disentangling the drivers of SIF and photosynthesis. We use Equations 5 and 6 in **Figure 3**, with a simplification of certain processes, to describe the factors driving SIF and photosynthesis (or GPP) at either the leaf or the canopy scale. The equations are intended to capture the first-order controls of SIF (W·m^{-2}·sr^{-1}) and GPP (μmol CO_2·m^{-2}·s^{-1}). We assume that leaf color (i.e., ω_{PAR}), Φ_{SIF}, and light-use efficiency (LUE) within the canopy are constant, and the impact of the soil background is ignored (168).

In Equation 5 in **Figure 3**, $i_0 \cdot (1 - \omega_{PAR}) \cdot 1/(1 - p \cdot \omega_{PAR})$ is essentially fPAR, the fraction (%) of PAR [photosynthetic photon flux density (PPFD); μmol photon·m^{-2}·s^{-1}] absorbed by the leaf or canopy (mainly by pigments[1]). We intentionally write it this way instead of using fPAR or absorbed PAR (APAR) to emphasize that at the canopy scale, fPAR is driven by both leaf-level processes ($1 - \omega_{PAR}$) and canopy-level processes (i_0 and p). The product of the first five terms in Equation 5 in **Figure 3** is the total SIF emission, and $f_{esc}(\lambda, \Omega)$, varying with wavelengths (λ) and viewing

[1]At the canopy scale, other nonphotosynthetically active components and soil background affect the amount of the absorbed PAR and thereby fPAR. In this case, fPAR absorbed by chlorophyll (fPARchl) is used in Equations 5 and 6 in **Figure 3**.

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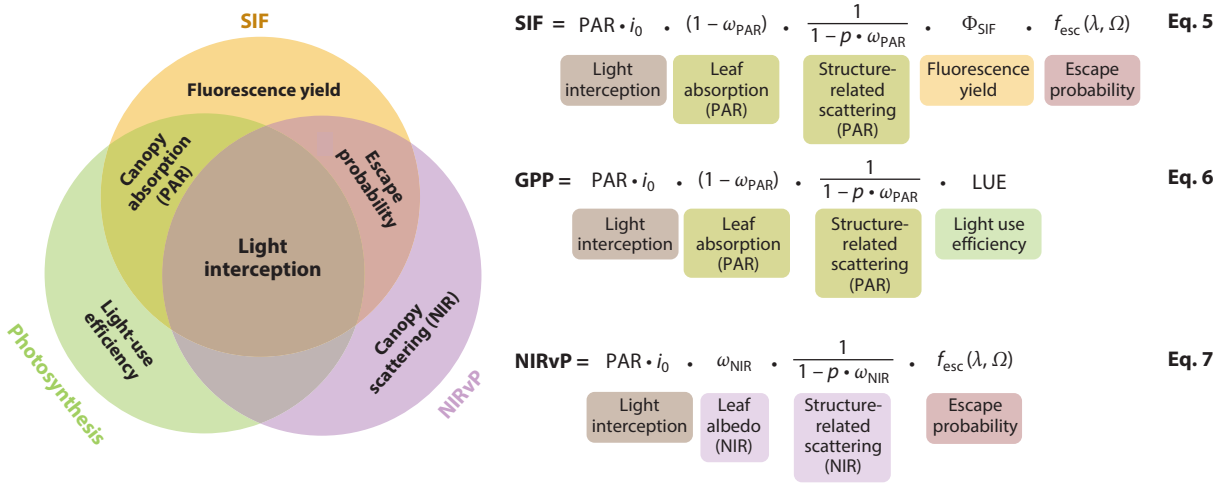


Figure 3

Conceptual diagram and key equations to understand the drivers and the relationships among SIF, GPP, and NIRvP. Equations assume a black soil scenario (no radiation reflected back to the canopy by soil). Canopy absorption (PAR) is the product of leaf absorption (PAR) and structure-related scattering (PAR). Canopy scattering (NIR) is the product of leaf albedo (NIR) and structure-related scattering (NIR). The relative areas of the overlap are semi-quantitative: Under normal environmental conditions, SIF yield varies between 1–5% of the absorbed PAR throughout the day. Light intercepted and absorbed by the canopy, however, may change by an order of magnitude throughout the day and thus contributes to >50–90% of diurnal variations in SIF radiances. The relative contributions of each component vary under different environmental and biological conditions and depend on the spatiotemporal scales. i_0 is light interceptance (either at the leaf or canopy scale); ω_{PAR} and ω_{NIR} are the leaf albedo (unitless, 0–1) in the PAR and NIR domains; p represents the probability (0–1) of a photon recolliding with the canopy after its initial nonabsorption interaction with the canopy (see **Figure 1*b***); and $f_{esc}(\lambda, \Omega)$ (unit: sr^{-1}) is the escape ratio, the percentage of total emitted fluorescence escaping a leaf or a canopy and detected by an optical device at a specific viewing angle. Abbreviations: Φ_{SIF}, SIF yield; LUE, light-use efficiency; PAR, photosynthetically active radiation.

angle (Ω), is a function of the leaf and canopy structure. At the canopy or ecosystem scales, Φ_{SIF} is the canopy fluorescence yield, collectively contributed by both PSI and PSII from leaves with different levels of biochemical traits (e.g., sunlit versus shaded leaves). $\Phi_{SIF} = [\alpha_2/\alpha \cdot \Phi_{F2} \cdot S_{F2}(\lambda) + \alpha_1/\alpha \cdot \Phi_{F1} \cdot S_{F1}(\lambda)] \cdot P_{esc}(\lambda, \Omega) = SIF(\lambda, \Omega)/(APAR \cdot f_{esc}(\lambda, \Omega))$ in which $P_{esc}(\lambda, \Omega)$ is photosystem-to-leaf escape ratio while $f_{esc}(\lambda, \Omega)$ is only related to canopy structure. $S_{F2}(\lambda)$ and $S_{F1}(\lambda)$ are the spectral shapes of fluorescence emission from PSII and PSI as a function of wavelength. Φ_{SIF} is also different from the term SIF yield, which has been used in some studies and which is effectively $\Phi_{SIF}^* f_{esc}(\lambda, \Omega)$ (e.g., 163). LUE ($\mu mol\ CO_2 \cdot m^{-2} \cdot s^{-1}/\mu mol\ photon \cdot m^{-2} \cdot s^{-1}$) is the efficiency to convert absorbed photons into carbon uptake, and it equals $\alpha_2/\alpha \cdot \Phi_{P2} \cdot k . \alpha_2/\alpha$ is the ratio between light absorbed by PSII and the total absorbed light by PSI and PSII, which is often assumed to be 0.5 despite its significant variability. k is the mole of CO_2 molecules reduced per mole of electron equivalents produced by linear electron transport.

5.1. How Is Solar-Induced Chlorophyll Fluorescence Related to Gross Primary Productivity?

The equations for SIF and GPP (Equations 5 and 6 in **Figure 3**) share the first four terms that describe the total PAR absorbed. Hence, we can combine these equations to write a single equation for GPP as a function of SIF and some other terms (Equation 8):

$$GPP = SIF \cdot \frac{LUE}{\Phi_{SIF} \cdot f_{esc}(\lambda, \Omega)}. \quad 8.$$

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The behavior of these additional terms is critical. If they are constant under various environmental conditions for all types of biomes, then GPP is proportional to SIF. Detailed consideration reveals that these other terms are not constant under all conditions, but we show that they are approximately constant under the conditions at which we currently measure satellite SIF. The other terms of Equations 5 and 6 in **Figure 3** are different and thus are further examined below.

At the leaf level, several studies have examined the relationship between Φ_F and photochemical yield (Φ_{P2}) and found that the relationship between Φ_F and Φ_{P2} varies by light intensity and the level of environmental stress (**Figure 4*a***). The level of NPQ plays a crucial role in the relationship between these two parameters (13, 24, 121). At low-light and low-stress conditions, NPQ is low and has not fully kicked in yet; the relationship between Φ_F and Φ_{P2} is negative as these two processes compete for the absorbed PAR. As light increases within an intermediate range (i.e., during most of the day), Φ_F and Φ_{P2} are positively correlated. At very high-light and high-stress conditions, this relationship may become negative again. However, more leaf-level measurements are needed to understand the relationship between Φ_F and Φ_{P2} (53, 67, 93). How biological drivers, such as leaf nitrogen content [and thus the maximum carboxylation rate of Rubisco (V_{cmax}) as well as the maximum activity of the Cytochrome b_6f complex (V_{qmax})], and environmental drivers affect this relationship remains to be tested across a wide range of conditions (144, 173).

Although the remaining parameters (f_{esc} and k) are not shared between SIF and GPP, many studies still show a robust linear relationship between SIF and GPP (37, 72, 88, 100, 163). The main reasons for this are twofold: (*a*) The structural factors that drive canopy light absorption in PAR (400–700 nm), mainly leaf area, leaf angle, clumping index (CI) (18), and leaf structural parameters such as leaf mass per area (LMA), are the same factors that also drive the scattering in the NIR domain (e.g., 800 nm) (108). Enhancing light absorption via multiple scattering in PAR, driven by leaf and canopy structure, also means boosting the multiple scattering of NIR radiation, effectively increasing the probability of NIR radiation escaping the canopy (130). Model simulations of canopy spectra have shown that increasing the leaf area index (LAI) results in an increasing NIR reflectance and a decreasing red reflectance (1). Thus, we would expect a positive relationship between $PAR \cdot i \cdot (1 - \omega_{PAR})$ and f_{esc} in the NIR domain. (*b*) Variations in k can result from shifts in the relative partitioning of electron transport in support of the photosynthetic carbon reduction (PCR) and photosynthetic carbon oxidation (PCO) cycles (67). For example, it is commonly observed that water limitation induces stomatal closure that restricts CO_2 supply, and in this situation, both k and Φ_{P2} decrease together. As a result, k and Φ_{P2} may covary over timescales ranging from minutes to days, as well as weeks to months. Similarly, over longer time scales, plants adjust canopy structures (e.g., leaf area or leaf angle), which drives the changes in absorbed radiation and f_{esc} (168). Overall, the collective impact of the physiological term (yields) and absorbed light (driven by light intensity and canopy structure) causes the relationship between SIF and GPP to become curvilinear or linear (22, 89, 94, 100, 159) (**Figure 4*b***). As we can see, GPP tends to saturate under high-light conditions, whereas SIF rarely does.

Equations 5 and 6 in **Figure 3** also provide information about the circumstances under which SIF may decouple from GPP. In short time scales (minutes to hours), in response to stress, leaves may close their stomata, causing a decrease in k, while all other parameters in the two equations change little. In this hypothetical situation, we would see a change in the relationship between SIF and GPP (57, 96). Similarly, the changes in the relationship between Φ_F and Φ_P could also result in a decoupling between SIF and GPP (75, 153). These decoupling scenarios provide insight into how plants function under stress.

Maximum activity of Cytochrome b_6f complex (V_{qmax}): a key parameter in plant physiology that quantifies the maximum catalytic activity of the Cytochrome b_6f complex, the primary rate-limiting factor in the photosynthetic electron transport system

Maximum carboxylation rate of Rubisco (V_{cmax}): a key parameter in plant physiology that quantifies the maximum carboxylase activity of the enzyme Rubisco, the primary rate-limiting factor in photosynthetic carbon metabolism

Leaf angle: leaf zenith angle is the angle between the leaf plane and the horizontal plane, and leaf azimuth angle is the angle formed clockwise from north to the horizontal projection of a leaf's axis

Clumping index (CI): describes the level of clumping between leaves and between individual plants; a higher CI corresponds to a lower level of clumping

Leaf area index (LAI): the one-sided leaf area per unit ground area (m^2/m^2)

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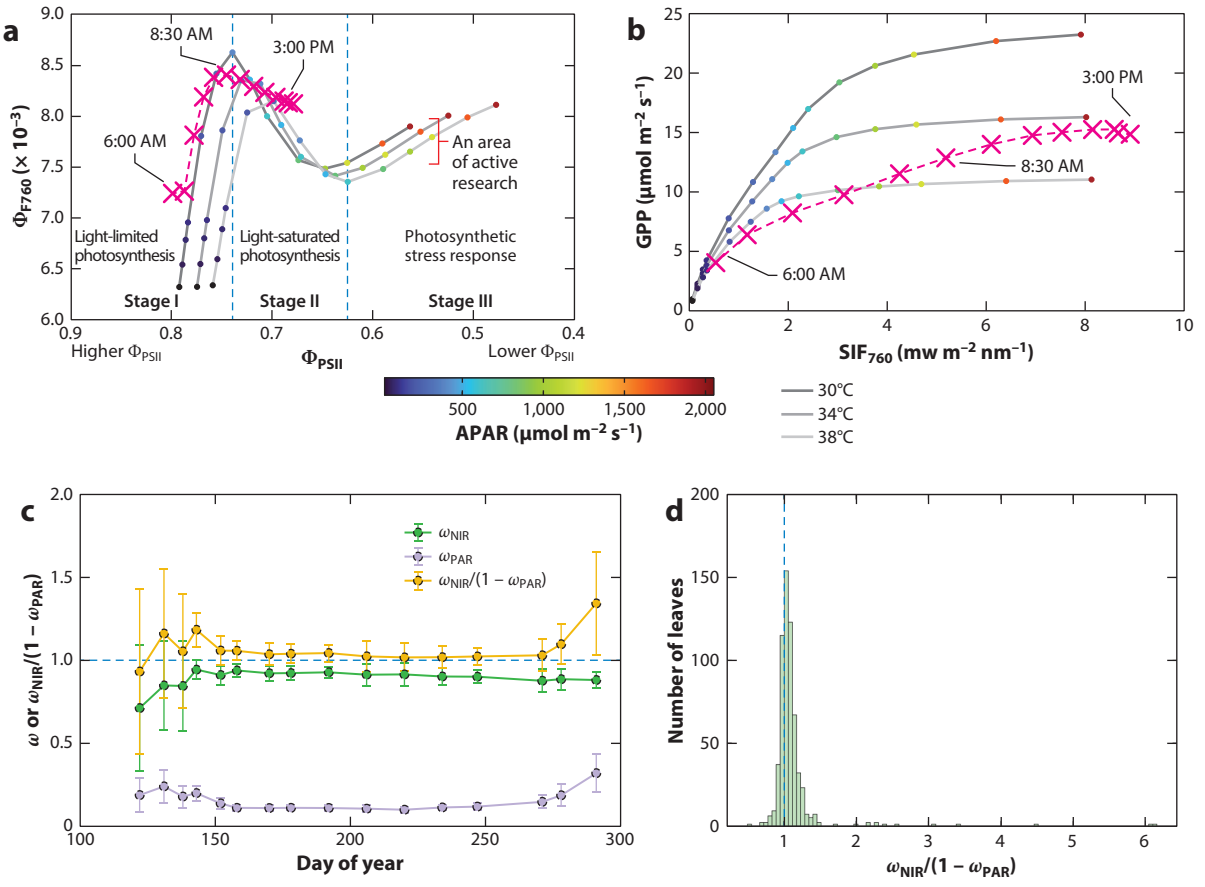


Figure 4

At the canopy scale, SIF is often proportional to GPP, even though the relationship between the yields of fluorescence and photosynthesis is nonlinear. The linearity between SIF and GPP is often observed in the range of light intensity, temperature, and time under which SIF measurements are made. The relationships between GPP and SIF, as well as Φ_{PSII} and Φ_{F760}, throughout a day (between 6 AM and 3 PM) are presented as magenta exes. Φ_{PSII} and Φ_{F760} are negatively correlated between 6 AM and 8:30 AM (note that the *x*-axis in panel *a* is inverted), and the relationship becomes positive between 8:30 AM and 3 PM. GPP and SIF are mostly proportional, but the slopes of the relationship change with light intensity. (*a*) Model-simulated relationship between the yield of PSII (Φ_{PSII}) and fluorescence yield at 760 nm (Φ_{F760}) across various PAR and temperatures (T_a) from 30°C to 38°C. Curves with different shades of gray are from simulations in which the air temperature was constant. The lighter the color of the curve, the higher T_a is. (*b*) Model-simulated relationship between SIF at 760 nm and GPP across various levels of PAR and T_a. Results are from SCOPE 2.1 (160). Details of model parameters can be found in the **Supplemental Appendix** and **Supplemental Figures 1** and **2**. Overall, the structure of the environmental response curves is driven by the properties of leaves that determine their photosynthetic capacity and then modulated by stress responses (67, 68). (*c*) The variations of the ratio between ω_{NIR} and $1 - \omega_{PAR}$ within a growing season in a temperate deciduous forest and (*d*) among mature leaves from various types of plants (33, 61, 164). Abbreviations: APAR, absorbed photosynthetically active radiation; GPP, gross primary productivity; NIR, near-infrared; PAR, photosynthetically active radiation; PSII, photosystem II; SIF, solar-induced chlorophyll fluorescence.

Supplemental Material >

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5.2. How Does Solar-Induced Chlorophyll Fluorescence Inform Us About Our Interpretation of Reflectance-Based Remote-Sensing Approaches for Photosynthesis?

Near-infrared reflectance from vegetation (NIRv): calculated as NDVI* NIR; other variations of NIRv include NIRvP (NIRv*PAR) and NIRv,Rad (NIRv* NIR radiance)

The unique aspect of SIF, compared with other remote-sensing metrics, is that it originates only from green vegetation. This is in contrast to reflectance-based approaches that see everything that reflects sunlight. For example, NDVI is sensitive to the mixture of green vegetation and other objects in a pixel (e.g., soil). This is commonly referred to as the mixed pixel problem. NIR reflectance from vegetation (NIRv) was developed to isolate the NIR reflectance from green vegetation in mixed pixels (6):

$$\mathrm{NIRv} = \rho_{\mathrm{NIR}} \cdot \mathrm{NDVI}. \qquad 9.$$

ρ_{NIR} is the NIR reflectance (0–1), and NDVI approximates the fraction of the measured ρ_{NIR} that originates from the green vegetation. At the global scale, NIRv is shown to be strongly correlated with SIF (6), and by extension of the arguments above, has become an alternative metric to estimate GPP (5, 21, 155). Note that NIRv is a reflectance (0–1), and SIF and GPP are fluxes. An alternative expression, NIRvP = NIRv · PAR (Equation 7 in **Figure 3**), has an even better correlation with SIF and GPP because it has a dimension of radiance and can be coupled to the incident PAR at the time of measurement (9). It is significant that these reflectance-based measurements, which should be primarily sensitive to canopy biophysics, capture much of the observed variation in SIF.

5.3. How Does the Relationship Between Near-Infrared Reflectance of Vegetation and Gross Primary Productivity Inform Us About the Role of Canopy Structure and Leaf Physiology?

Here, we delve into the ecological mechanisms underpinning the relationships between NIRv, GPP, and SIF. NIRvP shares several parameters with SIF, and we can use a similar approach to decompose NIRvP into four components (Equation 7 in **Figure 3**). In Equation 7, $1/(1-p\cdot\omega_{\mathrm{NIR}})$ is related to the multiscattering of NIR photons.

Combining Equations 6 and 7 (in **Figure 3**), with the assumption that $\omega_{\mathrm{NIR}} = 1$, $\omega_{\mathrm{PAR}} = 0$, and $(1-\omega_{\mathrm{PAR}})/(\omega_{\mathrm{NIR}}) = 1$ (**Figure 4*c***), we have the relationship between NIRvP and GPP:

$$\mathrm{GPP} = \mathrm{NIRvP} \cdot \frac{1-p}{f_{\mathrm{esc}}(\lambda, \Omega)} \cdot \mathrm{LUE}. \qquad 10.$$

Using the conventional LUE formulation, we can see that $\mathrm{NIRv}\cdot(1-p)/f_{\mathrm{esc}}$ is APAR. Canopy structure is a key factor affecting both the absorption and scattering of radiation in the canopy, thus affecting all three terms (NIRvP, SIF, and GPP), explaining some of the observed linear relationships among the terms (**Figure 3**). The observed linear relationship between GPP and NIRvP on the diurnal and seasonal scales (9, 21) suggests that the terms on the right-hand side of Equation 10 [i.e., $f_{\mathrm{esc}}(\lambda, \Omega)$, LUE, and p] either do not significantly change or covary with NIRvP. Indeed, NIRv and $f_{\mathrm{esc}}(\lambda, \Omega)$ likely covary seasonally (168). Under normal/unstressed conditions, LUE is a function of APAR (140). We can also see that when these conditions are not met, for example, under extreme drought, when k is close to zero while PAR is still high, or across different canopies with different $(1-\omega_{\mathrm{PAR}})/(\omega_{\mathrm{NIR}})$ ratios (**Figure 4*d***), the relationship between GPP and NIRvP may become nonlinear. The same is true for plants with different photosynthetic pathways: The relationship between GPP and NIRvP differs between C_3 and C_4 species, largely due to the differences in photorespiration (21).

Combining SIF and NIRvP, we can estimate fluorescence yield (Equation 11). When the ratio $(1-\omega_{\mathrm{PAR}})/(\omega_{\mathrm{NIR}})$ is close to 1 and does not vary significantly during the season, and multiple

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scattering is minimal ($p = 0$), SIF/NIRvP can be used as a proxy for Φ_F (169). However, high-quality SIF data are essential for this approach, as the small variation in Φ_F could be masked by noise in the SIF retrieval:

$$\Phi_{SIF} = \frac{\omega_{NIR}}{1 - \omega_{PAR}} \cdot \frac{SIF}{NIRvP} \rightarrow \Phi_{SIF} \propto \frac{SIF}{NIRvP}. \quad 11.$$

Leaf angle distribution (LAD): the statistical distribution of leaf angles in a plant canopy, which is a key parameter in radiative transfer models

5.4. Coordination Between Leaf and Canopy Traits Enhances the Relationship Between Solar-Induced Chlorophyll Fluorescence and Gross Primary Productivity

Trying to disentangle the impact of leaf physiology and canopy structure on SIF can be particularly challenging because the traits that control leaf physiology (e.g., pigment concentration) and the traits that control canopy structure may covary spatially and temporally (62, 63, 123, 162). For example, the physiological traits such as V_{cmax} and V_{qmax} that drive LUE and Φ_F at the leaf scale likely covary with the structural traits that drive f_{esc} at the canopy scale. Coordination among different plant traits has been a key theme in plant ecology (15). Certain trait combinations seem more evolutionarily stable and advantageous than other alternatives. This idea has developed into popular theories such as the leaf economics spectrum and the global spectrum of plant form and function (26, 151). As with the convergence of leaf traits, ecological and evolutionary optimality theory suggests a convergence between leaf traits and canopy structural traits (55); for example, one hypothesis is that canopies in natural ecosystems (e.g., noncrop types) with high nitrogen content (N%) have leaf angle distributions (LADs) that intercept a higher level of radiation, resulting in emergent ecosystem properties such as the relationship between canopy N% and the NIR albedo (79, 109, 162). Indeed, a recent study has shown that across ~320 eddy covariance sites globally, the maximum LUE and NIRv are tightly correlated (20).

Because the covarying structural traits and leaf traits control canopy reflectance and fluorescence, it becomes challenging to disentangle the structural and leaf physiological drivers of SIF (108, 162). Integrating remote-sensing metrics of canopy structure (such as NIRv) and SIF has been proposed to disentangle f_{esc} and Φ_F (168). This disentanglement is essential to understand the roles of structure and physiology in modulating observed fluorescence signals. This approach has been empirically demonstrated to be effective (169). A key assumption of using NIRv to estimate f_{esc} is that $\omega_{NIR} = 1$ spatially and temporally. On the other hand, leaf-level measurements suggest that ω_{NIR} is 0.7–0.9 (1, 164), suggesting that the assumption is likely violated (**Figure 4*c***). It is possible that some of the factors that drive escape probability (e.g., canopy structure) also covary with NIR reflectance, resulting in a good approximation of f_{esc} using NIRv.

5.5. A Conceptual Equation to Understand the Drivers of Solar-Induced Chlorophyll Fluorescence

Here, we provide a conceptual equation to help understand the drivers of SIF. SIF varies according to biological drivers (BIO) and environmental drivers (ENV), their interactions (BIO × ENV), and source-object-sensor geometry (GEOMETRY) (Equation 12).

$$SIF = f\left((BIO + ENV + (BIO \times ENV)) \times GEOMETRY\right). \quad 12.$$

Biological drivers can be further divided into functional (e.g., pigment concentration and leaf nitrogen content) and structural (e.g., LAI, leaf angle, clumping, and leaf anatomy) components that may covary spatially or temporally. Light distribution is a major factor driving within-canopy variability in SIF: Sunlit leaves emit SIF at rates several times higher than shaded leaves (118, 120). In addition, vegetation structure determines how the emitted SIF signal interacts with other

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components (e.g., other leaves, branches, and soil background), and thus how much of the total emitted SIF can be seen by a sensor. Environmental factors can exert direct controls (e.g., radiation, temperature, and CO_2) and indirect controls (e.g., soil moisture and vapor pressure deficit) on SIF. The interaction between biological and environmental drivers can be viewed as an indirect control. For example, soil nitrogen availability can affect leaf chlorophyll concentration as well as V_{cmax} and V_{qmax}, directly influencing the magnitude of SIF emission (17). An additional layer of complexity is related to the observations themselves—the source-object-sensor geometry (54, 170). The optical instruments we use only observe the target at one specific angle at a time, and normally at a specific solar zenith and azimuth angle. And the probability that SIF escapes the canopy to be detected is directionally dependent. Thus, the magnitude, shape, and temporal variations of the observed fluorescence signal are affected by the angle of the observation and the angle of the incoming radiation (e.g., the Sun), the latter of which is affected by the location of the source relative to the target and the amount of scattering between the source and the target (e.g., a sunny day versus a cloudy day).

Equation 12 can describe the drivers of many vegetation remote-sensing metrics, including optical reflectance metrics such as NDVI and NIRv (6, 155), canopy surface temperature from thermal remote sensing, and vegetation optical depth from microwave remote sensing.

Scale is a fundamental concept in ecology (85). Depending on the spatial and temporal scales, the drivers of the observed SIF and their relative importance vary, and they modulate the relationship between photosynthesis and SIF. Over longer timescales (e.g., weeks to months) and larger spatial scales (e.g., across biomes), vegetation structure tends to be the dominant control of SIF and GPP, leading to the strong linear relationship between these two.

6. SOLAR-INDUCED CHLOROPHYLL FLUORESCENCE FOR GLOBAL PHOTOSYNTHESIS RESEARCH

6.1. A Brief History of Optical Remote Sensing for Photosynthesis

Optical satellite remote sensing has evolved over the past five decades to track photosynthesis, driven by advances in sensor technology and ecological modeling (127) (**Figure 5**). Beginning in the 1970s with the first Earth-observing satellites (158), early efforts focused on VIs such as NDVI (139). These indices were proxies for canopy greenness, LAI, or fPAR (3, 106). Another VI, the photochemical reflectance index, is sensitive to changes in the xanthophyll cycle (40). However, VIs generally lack the direct physiological connection to photosynthesis: They capture certain structural or biochemical traits but not the dynamic light reactions or carbon assimilation at the heart of photosynthesis.

In the 1990s and 2000s, LUE models (104) using remote-sensing data were developed to integrate VIs with environmental data to estimate GPP (131). Major products such as MODIS GPP use VIs as inputs for fPAR, downregulating potential productivity with empirical environmental stress terms (125). While a significant move forward for global photosynthesis products, these models face significant uncertainties, as their parameterizations struggle to represent diverse plant responses to environmental stresses and extremes (133).

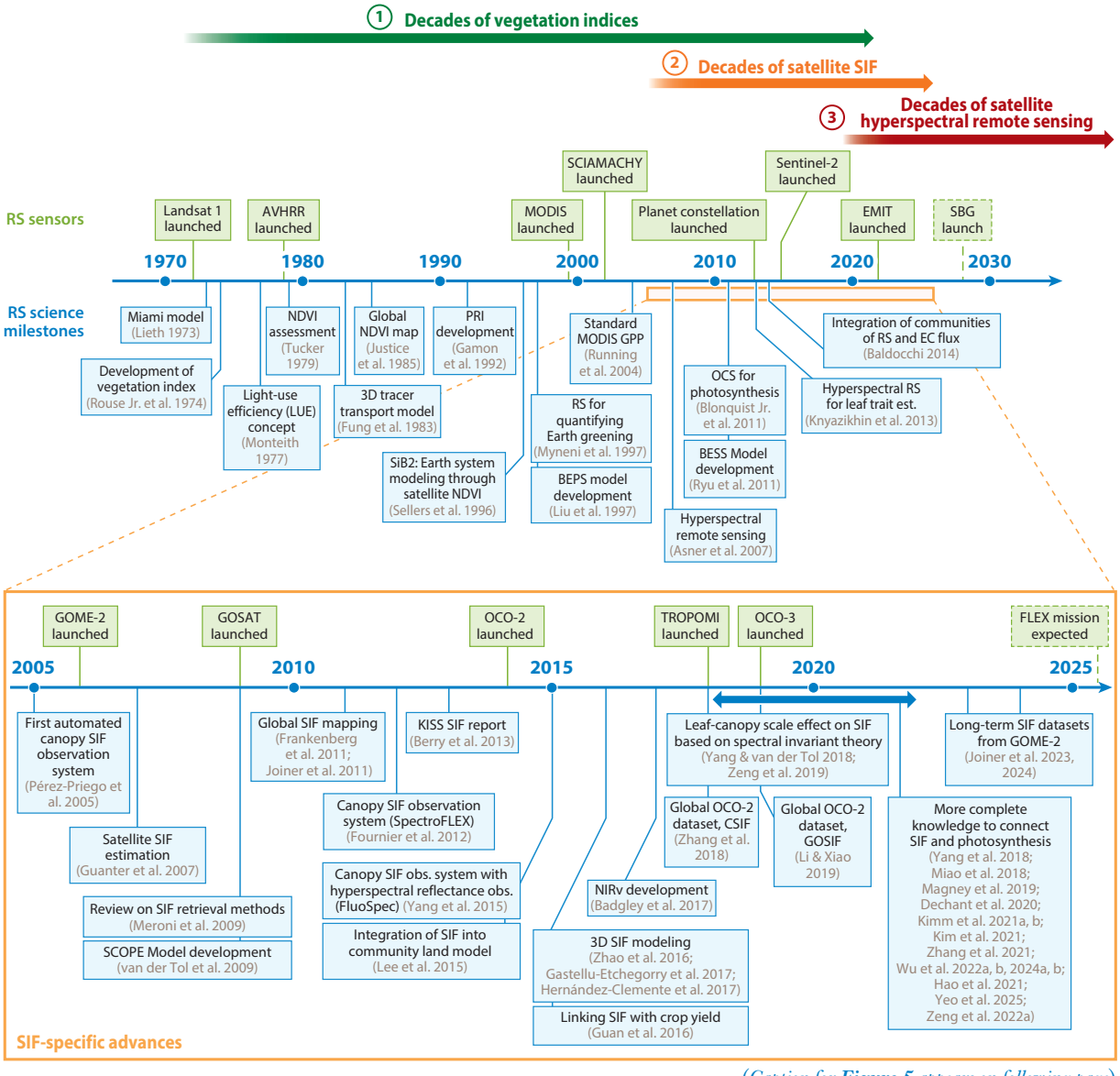
Since the first integration of the FvCB mechanistic photosynthesis model, in land surface models (i.e., SiB), models such as the Breathing Earth System Simulator (BESS), Biosphere-Atmosphere Exchange Process Simulator (BEPS), and Soil Canopy Observation of Photosynthesis and Energy Fluxes (SCOPE), along with other ecosystem models, have started to couple radiative transfer with the FvCB model (126, 142), enabling more robust links between satellite-derived vegetation variables and carboxylation kinetics of photosynthesis. Despite significant progress, these approaches still rely on indirect proxies of photosynthesis.

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The 2010s marked a turning point with the rise of satellite-based SIF. Global, continuous, and near-real-time observations significantly facilitated SIF and photosynthesis studies. Unlike VIs, SIF has a clear physical connection to photosystems (13): This justifies why SIF has been so prevalent in recent decades due to its transformative impact on sensing from vegetation color (e.g., via VIs) to photosynthetic activities. Regional and global SIF maps (**Supplemental Figure 1**) from GOSAT, GOME-2, OCO-2, and TROPOMI demonstrated robust correlations with GPP across biomes (37, 51, 69, 80, 136), validating its physiological basis (94). These sun-synchronous satellites, originally designed for greenhouse gas monitoring using very-high-spectral-resolution spectrometers, can provide coarse spatial resolution (greater than a few kilometers) and

Supplemental Material >

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(*Caption for* ***Figure 5*** *appears on following page*)

Figure 5 (*Figure appears on preceding page*)

A brief history of optical remote sensing for studying photosynthesis divided into three overlapping eras. (This list includes selective and representative work and is not meant to be comprehensive due to limited space.) (①) Decades of vegetation indices (1970s–present): Pioneered by sensors such as Landsat, AVHRR, and MODIS, this era established vegetation indices to monitor canopy greenness. (②) Decades of satellite SIF (2010s–present): Enabled by new satellite sensor technology, SIF emerged as a direct probe of photosynthetic light reactions. (③) Decades of satellite hyperspectral sensors (2020s–future): New satellite missions (e.g., EMIT, SBG, and FLEX) aim to unify canopy reflectance with SIF, resolving physiological and structural drivers of photosynthesis at high spatiotemporal scales. The zoomed panel highlights SIF-specific advances, including satellite sensors for SIF retrievals, canopy-scale SIF observations, and mechanistic SIF models, underscoring its transformative role in bridging spectral signals to photosynthetic function. Abbreviations: AVHRR, advanced very high resolution radiometer; EC, eddy covariance; EMIT, Earth surface mineral dust source investigation; FLEX, fluorescence explorer; GOSAT, greenhouse gases observing satellite; GPP, gross primary productivity; MODIS, moderate resolution imaging spectroradiometer; OCO-2, orbital carbon observatory-2; RS, remote sensing; SBG, surface biology geology; SCIAMACHY, scanning imaging absorption spectrometer for atmospheric cartography; SIF, solar-induced chlorophyll fluorescence; TROPOMI, tropospheric monitoring instrument. Timeline includes References 2, 6, 8, 13, 14, 22, 34, 37, 39, 40, 41, 48, 51, 54, 59, 70, 71, 72, 73, 75, 76, 77, 79, 82, 83, 87, 90, 94, 99, 100, 105, 106, 114, 124, 125, 126, 131, 139, 142, 152, 153, 156, 157, 159, 161, 163, 165, 168, 169, 174, 175, 176.

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inherently noisy datasets. Almost in parallel, there has been a growing deployment of tower-mounted SIF sensors, offering high-resolution, low-cost measurements that closely integrate with eddy covariance systems (34, 46, 75, 119, 150, 156, 159, 163), reveal spatial variability within flux footprints (154), support the development and validation of remote-sensing methods, and facilitate the scaling of site-level data to broader landscapes (66). In parallel, carbonyl sulfide (OCS) emerged as an independent tracer of gross primary production, offering complementary insights into photosynthetic carbon uptake due to its tight coupling with stomatal conductance and minimal reemission (14).

Beyond CO_2 uptake, SIF emerged as a versatile indicator of biological processes of photosynthesis, from photoprotective mechanisms (94) and phenology (50, 65) to crop productivity and yield (48, 52, 56). Another application is to study SIF as a plant stress indicator (57, 76, 77, 101, 152, 172). Recent modeling developments also suggest that it is possible to infer photosynthetic capacity using SIF observations (144, 173). Furthermore, SIF inspired NIRv (21, 155), which leverages canopy scattering properties to improve GPP estimates. Products such as SLOPE GPP (66), rooted in NIRvP, now rival or surpass SIF in accuracy, underscoring how SIF's foundational principles have permeated next-generation proxies.

Around the same period, advances in hyperspectral remote sensing provided new opportunities to capture vegetation function beyond traditional indices. Airborne systems combining hyperspectral imaging enabled the estimation of canopy biochemical and structural traits (2, 79). Global-scale hyperspectral remote-sensing observations have been absent until recent years despite being widely used at the leaf and ecosystem scales (1, 148, 164). In the past few years, new and future hyperspectral satellite missions, such as EnMap, PACE, and SBG, with global mapping capacity, have been launched or planned. They can be combined with SIF observations to disentangle the structural and functional drivers of SIF.

6.2. Modeling of Solar-Induced Chlorophyll Fluorescence at the Global Scale

Building on these advances in observation, there has been a surge in interest in modeling SIF, focusing on both physiological mechanisms and radiative transfer processes. On the physiological front, the traditional approach to modeling electron transport rate as an empirically derived function of light intensity has been replaced by new approaches relating SIF to photosynthesis (47, 67, 141). Regarding radiative transfer, models have evolved from simple big-leaf approaches to sophisticated three-dimensional representations (41, 59, 86, 89, 122, 128, 176). A prominent example that comprehensively integrates both physiological and radiative transfer components is

the SCOPE model (141, 142, 160). SCOPE combines radiative transfer, photosynthesis, and energy balance processes to simulate SIF emissions, offering an integrated framework for assessing SIF response to vegetation properties and environmental drivers.

Simulating SIF at the global scale allows for using satellite observations to constrain model simulations using data assimilation (107). Based on the framework of SCOPE, several major terrestrial biosphere models have incorporated simulations of SIF (4, 82, 84, 86, 107, 113, 147). These models generate similar simulations of Φ_F but show wide discrepancies at the canopy scale regarding total emitted fluorescence (113). Such discrepancies, while problematic for model intercomparisons, provide important information for understanding the often highly complex biochemical and physiological assumptions of the models.

Emerging artificial intelligence algorithms and process-guided machine learning facilitate novel model–data fusion techniques (49) that can leverage satellite SIF and photosynthesis data and reduce model uncertainties (91). These techniques enable model parameterization with satellite remote-sensing data in a spatially and temporally explicit manner. With the accelerated accumulation of data, the application of big-data-based causal inference approaches becomes increasingly promising. These approaches allow us to better understand and characterize the causal relationships among biogeophysical and biogeochemical variables. Ultimately, this will enhance our modeling strategies and approaches, providing improved accuracy and effectiveness.

7. CONCLUDING REMARKS AND FUTURE OUTLOOK

This review highlights the critical role of SIF in linking chloroplast-scale processes to global carbon fluxes and improving photosynthesis monitoring. SIF transcends its role as a proxy of photosynthesis. It reshaped our understanding of how chloroplast-scale biochemistry cascades to global-scale photosynthesis. In this review, we trace the evolution of remote-sensing approaches, from vegetation indices to SIF, while addressing challenges in scaling SIF from leaves to ecosystems. Understanding how canopy structure, radiative transfer, and observation geometry shape SIF signals is key to accurately interpreting SIF for photosynthesis estimation. We emphasize the need to disentangle physiological and structural drivers of SIF, examine its coupling and decoupling with photosynthesis across scales, and suggest that integrating SIF with the NIR reflectance of vegetation and process-based models may further improve large-scale GPP estimates and carbon cycle assessments. For future research on SIF, we recommend focusing on the following key scientific questions.

1. What are the ecophysiological mechanisms driving the steady-state fluorescence yield at the leaf level and its relationship with photochemical yield when plants are under various types of environmental stress? At shorter timescales (e.g., daily to weekly), studies show divergent results: Some found that SIF captures the physiological responses to stresses, while others show a decoupling of SIF from photosynthesis under stressed conditions (57, 97). The different responses might be due to the different types of stressors, varying stress levels, and/or different functional groups. More experimental studies are required to better understand the mechanisms of SIF and Φ_F for stress detection from the leaf to global scales.
2. To what extent and under what conditions does leaf physiology covary with canopy structure? Answering this question is essential for interpreting the spatiotemporal variation in SIF. SIF must be combined with other techniques: coupling canopy structural measurements from LiDAR or camera-based three-dimensional reconstruction, canopy biochemical traits from hyperspectral measurements, and SIF could help disentangle SIF signals (95, 117, 135, 171). Concurrent leaf-level PAM measurements are essential for interpreting the leaf physiological component (i.e., fluorescence yield) from SIF signals (116). Concurrent eddy

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covariance measurements allow for linking the light reactions with carbon metabolism (166). Integrating these measurements holistically will provide new insight to answer the question posed here.

3. What types of new instrumentation and observing strategies have the greatest potential to unlock the full power of SIF for Earth system science? To date, space-based SIF observations have been entirely opportunistic as the satellite missions were not designed for this target. Looking forward, new perspectives are anticipated from several current and upcoming missions: the European Space Agency's FLEX (28), which is the first mission to prioritize SIF; the National Aeronautics and Space Administration's (NASA's) TEMPO (177) and GeoCarb (132), which are the first missions with potential for retrieving terrestrial SIF from a geostationary orbit; and the joint Agenzia Spaziale Italiana–NASA mission Luce (formerly CALIGOLA) (12), which is the first mission designed to probe active fluorescence from space. There is great scope for further innovations in instrument design to optimize SIF measurement (e.g., by improving the spectral range, spectral resolution, and spatial resolution), as well as for innovations in observing strategy to optimize SIF interpretation (e.g., by placing additional instruments in geostationary orbits to complement those already in polar orbits).
4. Based on the insights from the above questions, how can we more effectively use SIF to improve photosynthesis estimates and model simulations across varying temporal and spatial scales? One promising direction is to move toward integrating multisensor remote sensing, mechanistic modeling, and AI-driven techniques. Considering the noticeable uncertainties in existing models, AI-driven modeling, especially the knowledge-guided machine learning techniques (91, 146), may help reduce such uncertainties and refine SIF-based GPP predictions.

Looking ahead, global-scale mechanistic understanding and simulation of photosynthesis are on the horizon (67). Achieving these goals requires communication and coordination among scientists from diverse fields, including but not limited to plant biologists, ecophysiologists, global change ecologists, and remote-sensing scientists. Combining approaches such as leaf gas exchange and fluorescence measurements, leaf and canopy spectroscopy, passive and active fluorimetry, and radiative transfer and photosynthesis modeling would allow greater progress on the topic. To build the observational foundation for improvements in biosphere model structure and parameterization, we need new controlled experiments providing insights to interpret observations in nature, as well as new instrumentation providing optimal characterization of SIF at the spatiotemporal scales relevant for biosphere modeling. These efforts will push the knowledge boundary of plant biology and make significant contributions to broader science fields such as global ecology and climate sciences.

SUMMARY POINTS

1. Solar-induced chlorophyll fluorescence (SIF) provides a remote-sensing signal that is linked explicitly to photosynthesis. Simultaneous studies of reflected light and SIF permit us to study how physiology and structure are interlinked to drive photosynthesis.
2. The spatiotemporal scale is a fundamental concept in understanding the relationship between SIF and photosynthesis. Biological and environmental factors, the interactions between these two types of factors, and viewing geometry all affect the relationship between SIF and photosynthesis.

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3. Active fluorescence techniques, such as pulse-amplitude modulation (PAM), have differences and similarities to passive fluorescence techniques, such as SIF. SIF yield (Φ_{SIF}) is conceptually similar to the steady-state fluorescence yield measured by PAM; thus, Φ_{SIF} gives a way to compare photon processing at the canopy scale. Studying fluorescence at small scales provides the ecophysiological understanding needed to interpret the biological, environmental, and observational controls on SIF at larger scales.
4. With current observation platforms, the relationship between SIF and photosynthesis tends to become more linear when moving from smaller to larger spatiotemporal scales.
5. To fully exploit the potential of SIF for Earth system science, there is a need to develop new instrumentation that is optimized for the measurement and interpretation of SIF across scales.

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AUTHOR CONTRIBUTIONS

X.Y., K.G., M.C., J.E.J., and J.A.B. conceived the framework. X.Y. wrote the first draft, conducted the data analysis, and drew **Figures 1–4**. K.G., G.W., and H.K. drew **Figure 5**. J.E.J. wrote Section 4 and edited the manuscript. All authors contributed to the writing of the manuscript.

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